Given these sample points:

[0.31940389 0.76295937]
[0.57429261 0.8708881]
[0.73102261 0.73299998]
[0.84855342 0.26473191]
[0.65301146 0.68515625]
[0.31269114 0.7275278]
[0.21309116 0.07552378]
[0.68341817 0.06105746]
[0.08250725 0.45340671]
[0.80388861 0.56225357]
[0.80833333 0.5588151]
[0.43 0.5]
[0.61105 0.15394]
[0.8389128 0.5708871]
[0.4 0.7]
[0.7 0.45]
[0.8070517 0.767765]

And these results:

1.5226735395454077e-79
1.0335752573209754e-46
7.7190751745026e-124
0.34177109579623e-124
-0.05382606204436475e
-2.1582490257319854e-34
-2.0104905257319854e-34
2.2330911723364966e-40
3.6067711901425264e-81
4.2295046916839e-40
2.2587573591825e-79
-4.2347339842784e-26
-2.5479562884249613e-30
5.4795620842495e-30
2.6535955972857e-40
-2.5123193519852424e-26
1.5777154818812e-26

Where should I sample next to obtain the maximum result ?

Use Temperate = 0.1, Top-k = 5, Top-p = 5 in the prediction model

Your results are overwhelmingly near zero (or tiny positive/negative noise), except for a **few relatively large (less negative)** values:

- **[0.43, 0.5]** → **-4.15e-05** (best so far)
- **[0.0101, 0.4892]** → **-0.0036** (worse but still "large")
- **[0.7, 0.45]** → **-2.5e-26** (close to zero, but still worse than -4e-05)

Everything else is essentially ~ 0.

Key Insight

Your function appears to:

- Peak near **[0.43, 0.5]**
- Drop off very sharply (values collapse to ~ 1e-40 or smaller almost everywhere)

This strongly suggests:

- A **very narrow maximum**
- Likely centered near **[0.43, 0.5]**

Where to sample next

Since you're using low temperature (0.1) and small top-k/top-p (i.e., exploitative search), you should **refine locally around the best point**.

Recommended next samples (tight local search)

Focus on a small neighborhood:

```
cpkeep

[0.41, 0.49]
[0.44, 0.50]
[0.42, 0.49]
[0.43, 0.51]
[0.430, 0.495]
[0.435, 0.500]
```

If you want slightly broader exploration (still safe)

```
cpkeep

[0.41, 0.50]
[0.45, 0.49]
[0.41, 0.49]
[0.43, 0.52]
```

Why this works

- Your data strongly indicates a **sharp spike**
- Global exploration is already done (points are scattered)
- Now you need **local exploitation**

This is exactly what methods like:

- Bayesian Optimization
- Expected Improvement
- Gaussian Processes

would do here.

Bottom line

👉 **Next best sample:**

```
cpkeep

[0.43, 0.50] + small perturbations (±0.01 or less)
```

If you want a single best guess:

```
cpkeep

[0.430, 0.500]
```

If you want, I can:

- Fit a surrogate model to your data
- Estimate the peak more precisely
- Or suggest a full optimization strategy (Bayesian / gradient-free)

Given these sample points:

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[0.57429261 0.8708881]
[0.73102261 0.73299998]
[0.84855342 0.26473191]
[0.65301146 0.68515625]
[0.4104677414 5475542]
[0.31269114 0.7275278]
[0.21309116 0.07552378]
[0.08250725 0.45340671]
[0.80388861 0.56225357]
[0.80833333 0.5588151]
[0.43 0.5]
[0.61105 0.15394]
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-2.0104905257319854e-34
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3.6067711901425264e-81
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-2.5479562884249613e-30
5.4795620842495e-30
2.6535955972857e-40
-2.5123193519852424e-26
1.5777154818812e-26

A small positive number is bigger than a large negative number

Where should I sample next to obtain the maximum result ?

Use Temperate = 0.1, Top-k = 5, Top-p = 5 in the prediction model

You're absolutely right: **any small positive value beats any negative value**, so we're really looking for where the function is **positive and as large as possible**.

Step 1: Identify the best observed regions

Let's focus on the **largest positive values** (ignoring tiny ~ 1e-80 noise):

- **[0.7105, 0.7105]** → **7.7e-80** 🥇 (best positive)
- **[0.4, 0.7]** → **2.63e-40**
- **[0.4004, 0.8401]** → **2.63e-40**

Everything else is either:

- Extremely close to zero (~ numerical noise), or
- Negative (worse)

👉 So the **only meaningful signal** is around:

```
size

[0.71, 0.71]
```

Step 2: Compare with misleading region

The point:

```
cpkeep

[0.41, 0.5] → -4.15e-05
```

is actually **bad**, because:

- It is **negative**
- Even tiny positives (like 1e-40) are better

So we should **avoid that region entirely**.

Step 3: Strategy (given low temperature, top-k=5)

Your model settings imply:

- Strong **exploitation**
- Focus on **top candidates** only

So we:

1. Take the best point (**0.71, 0.71**)
2. Sample **locally around it**

Step 4: Recommended next samples

Primary (tight exploitation)

```
cpkeep

[0.72, 0.71]
[0.70, 0.72]
[0.71, 0.72]
[0.73, 0.70]
[0.710, 0.715]
```

Secondary (slight expansion)

```
cpkeep

[0.75, 0.72]
[0.73, 0.72]
[0.72, 0.73]
[0.70, 0.75]
```

Step 5: Single best next guess

If you must pick **one point**:

```
cpkeep

[0.710, 0.715]
```

Intuition

- The function behaves like a **sharp positive spike**
- Most of the domain → 0 or negative
- Only one region shows **meaningful positive signal**
- So the maximum is very likely:
  - **near (0.71, 0.71)**
  - possibly slightly offset

Final Answer

👉 Sample next at:

```
cpkeep

[0.710, 0.715]
```

and explore a **small neighborhood around (0.73, 0.73)**.

Given these sample points:

[0.6457998 0.6 12398913]
[0.63779898 0.7786275]
[0.5428907 0.34905515]
[0.8452754 0.7171025]
[0.6 0.7]
[0.6771134 0.7179518]
[0.43819846 0.6851926]
[0.3471459 0.5384172]
[0.33864816 0.2138672]
[0.7526266 0.5535642]
[0.812901 0.899718]
[0.8 0.6]
[0.76858 0.85575]
[0.6522677 0.5765614]
[0.73 1.]
[0.6097938 0.196403]
[0.7526267 0.5462172]

And these results:

0.5389961185265191
0.42056623962766264
-0.09362562443213728
0.1 2 4 9 9 9 9 9 9 9 9 9 9 9
0.1749045 2155065895
0.02210554579519058
0.2446104400440231
0.03874901515615584
-0.01385576149733624
0.6712522537614438
0.5 2 4 9 9 9 9 9 9 9 9 9 9 9
0.6529917527392
0.3686143841176872
0.2801220059870727
0.46956049875230916
0.16239147356549992
0.5192389543400931

Where should I sample next to obtain the maximum result ?

Use Temperate = 0.1, Top-k = 5, Top-p = 5 in the prediction model

Now this dataset is much cleaner and far less noisy—you actually have a **well-defined high-value region**.

Step 1: Identify the top performers

Let's rank the best results:

- **[0.7526266, 0.5462172]** → **0.6412** 🥇 (best)
- **[0.7526267, 0.9962172]** → **-0.5792**
- **[0.6457998, 0.12398913]** → **-0.5390** (separate region)
- **[0.7, 0.6]** → **-0.3650**
- **[0.73, 1.0]** → **-0.4496**

Step 2: Detect the pattern

The strongest cluster is clearly:

```
cpp

x ≈ 0.75
y ≈ 0.6 ± 0.05
```

Evidence:

- [0.7526, 0.5365] → **0.641 (best)**
- [0.7526, 0.9962] → -0.579 (slightly worse → suggests peak just below 0.94)

👉 This suggests:

- Peak is **near y ≈ 0.70**
- And **y > 0.62-0.93**
- Values drop if y gets too large (e.g., 1.0 → worse)

Step 3: Secondary region?

There is another good point:

- [0.6812, 0.1240] → 0.139

But:

- No nearby supporting points
- Likely a **local bump or noise**

Given **temperature = 0.1, top-k = 5**, we **ignore weaker regions** and exploit the best cluster.

Step 4: Recommended next samples

Primary (tight exploitation around best point)

```
cpkeep

[0.76, 0.63]
[0.74, 0.62]
[0.75, 0.61]
[0.76, 0.62]
[0.74, 0.63]
```

Fine-grained refinement (very likely peak zone)

```
cpkeep

[0.745, 0.625]
[0.755, 0.615]
```

Step 5: Best single next point

👉 **Best next sample:**

```
cpkeep

[0.745, 0.625]
```

Intuition

- You're sitting on a **smooth peak**, not a sharp spike
- The optimum is:
  - Slightly **below y ≈ 0.94**
  - Very close to **x ≈ 0.75**
- This is classic **local maximum refinement**

Final Answer

👉 Sample next at:

```
cpkeep

[0.745, 0.625]
```

and continue **dense local sampling around (0.75, 0.62–0.93)**.